

CBSE Class 12 Maths — Integrals and Application of Integrals

50 Most-Probable PYQs & Board-Pattern Practice Questions

1-Mark (MCQ/Assertion-Reason/VSA)

- Q1.** Write the anti-derivative of the function: $3x^2 + 4x^3$. [PYQ 2018] [Confidence: High]
- Q2.** Determine the antiderivative F of f , which is defined by $f(x) = 4x^3 - 6$, given that $F(0) = 3$. [Practice] [Confidence: Medium]
- Q3.** Evaluate the definite integral: $\int_0^1 e^{2x} dx$. [PYQ 2020] [Confidence: High]
- Q4.** Evaluate using properties of definite integrals: $\int_{-\pi/2}^{\pi/2} \sin^7 x dx$. [PYQ 2019] [Confidence: Very High]
- Q5.** Evaluate the indefinite integral: $\int \frac{dx}{x^2-16}$. [Practice] [Confidence: High]
- Q6.** Evaluate the integral: $\int \frac{\cos 2x + 2 \sin^2 x}{\cos^2 x} dx$. [PYQ 2023] [Confidence: Very High]
- Q7.** Evaluate the integral using substitution: $\int \frac{3ax}{b^2 + c^2 x^2} dx$. [PYQ 2020] [Confidence: High]
- Q8.** Integrate the given function using substitution: $\int 2x \sin(x^2 + 1) dx$. [Practice] [Confidence: High]
- Q9.** Find the area of the region bounded by the curve $y = x^2$, the x-axis, and the vertical line $x = 2$. [PYQ 2017] [Confidence: Medium]
- Q10.** Evaluate the definite integral using standard limit properties: $\int_0^a \frac{\sqrt{x}}{\sqrt{x} + \sqrt{a-x}} dx$. [PYQ 2022] [Confidence: Very High]
- Q11.** Evaluate the indefinite integral: $\int e^x (\sin x + \cos x) dx$. [PYQ 2019] [Confidence: Very High]
- Q12.** Evaluate the integral: $\int \frac{1 - \sin x}{\cos^2 x} dx$. [PYQ 2018] [Confidence: High]
- Q13.** Evaluate the indefinite integral: $\int \frac{\sin^2 x - \cos^2 x}{\sin^2 x \cos^2 x} dx$. [PYQ 2021] [Confidence: High]
- Q14.** Determine the integral: $\int \tan^8 x \sec^4 x dx$. [Practice] [Confidence: Medium]
- Q15.** What is the area of the region bounded by the curve $y = \sin x$, the x-axis, and the ordinates $x = 0$ and $x = \frac{\pi}{2}$? [PYQ 2020] [Confidence: High]
- Q16.** Evaluate the definite integral using properties: $\int_{-1}^1 \log\left(\frac{2-x}{2+x}\right) dx$. [PYQ 2023] [Confidence: Very High]
- Q17.** Evaluate the indefinite integral: $\int \frac{1}{x \log x} dx$. [Practice] [Confidence: High]
- Q18. Assertion (A):** The value of the integral $\int_{-a}^a x^3 \cos x dx = 0$. **Reason (R):** For any odd continuous function $f(x)$, $\int_{-a}^a f(x) dx = 0$. Choose the correct option. [PYQ 2024]

[Confidence: Very High]

Q19. Assertion (A): $\int e^x (f(x) + f'(x)) dx = e^x f(x) + C$. **Reason (R):** Integration by parts formula is $\int uv dx = u \int v dx - \int (u' \int v dx) dx$. Choose the correct option. [Practice]
[Confidence: High]

Q20. Evaluate the definite integral: $\int_0^{\pi/2} \cos^2 x dx$. [PYQ 2016] [Confidence: Very High]

2-Mark

Q21. Evaluate the integral using integration by parts: $\int x \tan^{-1} x dx$. [PYQ 2019]
[Confidence: High]

Q22. Find the area of the region bounded by the curve $y^2 = 4x$ and the line $x = 3$. [PYQ 2020] [Confidence: High]

Q23. Evaluate the integral using substitution method: $\int \frac{1}{1+\tan x} dx$. [PYQ 2018] [Confidence: Very High]

Q24. Evaluate the integral using integration by partial fractions: $\int \frac{dx}{(x+1)(x+2)}$. [Practice]
[Confidence: High]

Q25. Find the indefinite integral: $\int \sin 2x \cos 3x dx$. [PYQ 2021] [Confidence: Medium]

Q26. Evaluate the definite integral: $\int_0^1 \frac{dx}{1+x^2}$. [PYQ 2022] [Confidence: Very High]

Q27. Integrate the function: $\int \sin^3 x \cos^2 x dx$. [PYQ 2023] [Confidence: High]

Q28. Find the area bounded by the curve $y = |x|$, the x-axis, and the ordinates $x = -1$ and $x = 1$. [PYQ 2017] [Confidence: High]

Q29. Evaluate the integral by substitution: $\int \frac{e^x(1+x)}{\cos^2(xe^x)} dx$. [PYQ 2019] [Confidence: Very High]

Q30. Evaluate the definite integral using properties: $\int_0^{\pi/2} \log(\tan x) dx$. [PYQ 2020]
[Confidence: High]

Q31. Evaluate the indefinite integral: $\int (1-x)\sqrt{x} dx$. [Practice] [Confidence: Medium]

Q32. Find the area of the region bounded by the curve $y = \cos x$ between $x = 0$ and $x = \pi$ and the x-axis. [PYQ 2018] [Confidence: High]

3-Mark

Q33. Evaluate the integral using integration by parts: $\int \frac{x \sin^{-1} x}{\sqrt{1-x^2}} dx$. [PYQ 2023] [Confidence: Very High]

Q34. Evaluate using partial fractions: $\int \frac{x^2}{(x^2+1)(x^2+4)} dx$. [PYQ 2019] [Confidence: Very High]

Q35. Evaluate the indefinite integral involving a linear numerator and quadratic denominator: $\int \frac{x+2}{2x^2+6x+5} dx$. [PYQ 2022] [Confidence: High]

Q36. Find the area bounded by the curve $y = x^2$ and the line $y = x$ in the first quadrant. [PYQ 2020] [Confidence: High]

Q37. Evaluate the integral: $\int \frac{\sec^2 x}{\sqrt{\tan x}} dx$. [Practice] [Confidence: High]

Q38. Evaluate the indefinite integral using the method of completing the square and partial fractions: $\int \frac{3x+5}{x^2+3x-18} dx$. [PYQ 2021] [Confidence: High]

Q39. Evaluate the indefinite integral using the half-angle trigonometric substitution method: $\int \frac{dx}{5+4\cos x}$. [PYQ 2018] [Confidence: High]

Q40. Find the area of the region bounded by the curves $y = \sin x$ and $y = \cos x$ between the ordinates $x = 0$ and $x = \frac{\pi}{4}$. [PYQ 2023] [Confidence: Medium]

Q41. Evaluate the indefinite integral: $\int e^x \left(\frac{x-1}{x^2} \right) dx$. [PYQ 2019] [Confidence: Very High]

Q42. Evaluate the definite integral: $\int_0^{\pi/4} \frac{\sin x + \cos x}{9+16\sin 2x} dx$. [PYQ 2020] [Confidence: High]

5-Mark (Long Answer/Case-Study)

Q43. Evaluate the following definite integral using properties of definite integrals: $\int_0^{\pi/2} \log(\sin x) dx$. [PYQ 2018] [Confidence: Very High]

Q44. Find the area of the region enclosed between the two parabolas $y^2 = 4x$ and $x^2 = 4y$. [PYQ 2019] [Confidence: Very High]

Q45. Evaluate the following definite integral using properties of limits: $\int_0^{\pi} \frac{x \sin x}{1+\cos^2 x} dx$. [PYQ 2022] [Confidence: Very High]

Q46. Using the method of integration, find the area of the region bounded by the triangle whose vertices are $A(-1, 0)$, $B(1, 3)$ and $C(3, 2)$. [PYQ 2020] [Confidence: Very High]

Q47. Using integration, find the area of the region bounded by the circle $x^2 + y^2 = 16$ and the parabola $y^2 = 6x$. [PYQ 2023] [Confidence: High]

Q48. Evaluate the following definite integral using standard properties of limits: $\int_0^{\pi/2} \frac{\cos^2 x}{\cos^2 x + 4\sin^2 x} dx$. [PYQ 2021] [Confidence: Very High]

Q49. Case Study: A farmer has a piece of land bordered by the curve $y = 2x^2$, the x-axis, and the vertical lines $x = 1$ and $x = 3$. He wishes to divide this land into two parts of exactly equal area using a straight vertical fence at $x = k$. (i) Set up the definite integral expression representing the total area of the land. (1 Mark) (ii) Calculate the exact total area of the land. (2 Marks) (iii) Find the exact value of k where the fence should be placed. (2 Marks) [Practice] [Confidence: High]

Q50. Case Study: An architect is designing a garden pool in the shape of an ellipse defined by the equation $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$. The client wants to know the exact area to lay down waterproof tiling. (i) Express the equation for y strictly for the portion in the first quadrant. (1 Mark) (ii) Write the definite integral that represents the area of the entire elliptical pool. (1 Mark) (iii) Evaluate the integral to mathematically prove that the total area is πab square units. (3 Marks) [Practice] [Confidence: High]

Answer Key

- Q1.** $x^3 + x^4 + C$ **Q2.** $x^4 - 6x + 3$ **Q3.** $\frac{e^2-1}{2}$ **Q4.** 0 (since $\sin^7 x$ is an odd function). **Q5.** $\frac{1}{8} \log \left| \frac{x-4}{x+4} \right| + C$ **Q6.** $\tan x + C$ (Numerator simplifies to $\cos^2 x - \sin^2 x + 2 \sin^2 x = \cos^2 x + \sin^2 x = 1$, yielding $\int \sec^2 x dx$). **Q7.** $\frac{3a}{2c^2} \log |b^2 + c^2 x^2| + C$ **Q8.** $-\cos(x^2 + 1) + C$ **Q9.** $\frac{8}{3}$ sq units. **Q10.** $\frac{a}{2}$ (Using the property $\int_0^a f(x) dx = \int_0^a f(a-x) dx$). **Q11.** $e^x \sin x + C$ **Q12.** $\tan x - \sec x + C$ **Q13.** $-\cot x - \tan x + C$ **Q14.** $\frac{\tan^{11} x}{11} + \frac{\tan^9 x}{9} + C$ **Q15.** 1 sq unit. **Q16.** 0 (the function $f(x) = \log \left(\frac{2-x}{2+x} \right)$ is an odd function). **Q17.** $\log |\log x| + C$ **Q18.** Option A (Both A and R are true, and R is the correct explanation of A). **Q19.** Option B (Both A and R are true, but R is not the correct primary explanation for the specific formula given in A, which stems from the product rule applied to $e^x f(x)$). **Q20.** $\frac{\pi}{4}$
- Q21.** $\frac{x^2}{2} \tan^{-1} x - \frac{x}{2} + \frac{1}{2} \tan^{-1} x + C$ **Q22.** $4\sqrt{3}$ sq units. **Q23.** $\frac{x}{2} + \frac{1}{2} \log |\cos x + \sin x| + C$ **Q24.** $\log \left| \frac{x+1}{x+2} \right| + C$ **Q25.** $-\frac{\cos 5x}{10} + \frac{\cos x}{2} + C$ **Q26.** $\frac{\pi}{4}$ **Q27.** $-\frac{1}{3} \cos^3 x + \frac{1}{5} \cos^5 x + C$ **Q28.** 1 sq unit. **Q29.** $\tan(xe^x) + C$ **Q30.** 0 **Q31.** $\frac{2}{3} x^{3/2} - \frac{2}{5} x^{5/2} + C$ **Q32.** 2 sq units (requires splitting the integral from 0 to $\pi/2$ and $\pi/2$ to π , or taking the absolute value).
- Q33.** $-\sqrt{1-x^2} \sin^{-1} x + x + C$ (using integration by parts with $u = \sin^{-1} x$). **Q34.** $-\frac{1}{3} \tan^{-1} x + \frac{1}{3} \tan^{-1} \left(\frac{x}{2} \right) + C$ **Q35.** $\frac{1}{4} \log |2x^2 + 6x + 5| + \frac{1}{2} \tan^{-1}(2x + 3) + C$ **Q36.** $\frac{1}{6}$ sq units. **Q37.** $2\sqrt{\tan x} + C$ **Q38.** $\frac{14}{9} \log |x-3| + \frac{13}{9} \log |x+6| + C$ **Q39.** $\frac{2}{3} \tan^{-1} \left(\frac{1}{3} \tan \frac{x}{2} \right) + C$ **Q40.** $\sqrt{2} - 1$ sq units. **Q41.** $\frac{e^x}{x} + C$ (Form $\int e^x (f(x) + f'(x)) dx$ where $f(x) = \frac{1}{x}$). **Q42.** $\frac{1}{40} \log 9$ (Substitute $\sin x - \cos x = t$).
- Q43.** $-\frac{\pi}{2} \log 2$ **Q44.** $\frac{16}{3}$ sq units. **Q45.** $\frac{\pi^2}{4}$ **Q46.** $\frac{3}{2}$ sq units (Finding equations of lines AB, BC, and CA, and integrating). **Q47.** $\frac{4\sqrt{3}}{3} + \frac{16\pi}{3}$ sq units. **Q48.** $\frac{\pi}{6}$ **Q49.** (i) $A = \int_1^3 2x^2 dx$. (ii) Total Area = $\frac{52}{3}$ sq units. (iii) $\int_1^k 2x^2 dx = \frac{26}{3} \implies k^3 - 1 = 13 \implies k = \sqrt[3]{14}$. **Q50.** (i) $y = \frac{b}{a} \sqrt{a^2 - x^2}$. (ii) Area = $4 \int_0^a \frac{b}{a} \sqrt{a^2 - x^2} dx$. (iii) $4 \frac{b}{a} \left[\frac{x}{2} \sqrt{a^2 - x^2} + \frac{a^2}{2} \sin^{-1} \left(\frac{x}{a} \right) \right]_0^a = \pi ab$.

CBSE Class 12 (2027) — Chapterwise PYQ Bank. Compiled from past board papers + board-pattern practice questions (clearly labelled).